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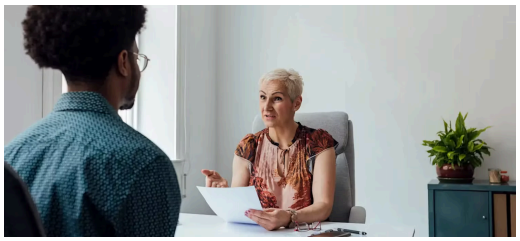
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Software Engineer Job Description: A Detailed Guide

Written by Coursera Staff • Updated on May 1, 2026
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Software engineers build, test, and maintain software applications for a variety of business cases and end users. As digital technology continues to evolve, the demand for software engineers will grow. Learn about the software engineer job description.



Key takeaways

Software engineers design, construct, test, and maintain software applications for various business use cases and end users.

- Software engineers typically need skills in coding, debugging, object-oriented programming, database management, and communication.

- Some roles you could pursue in software engineering include front-end developers, back-end developers, full-stack developers, mobile-app developers, and DevOps.
- You can become a software engineer by earning a bachelor's degree in software engineering, computer science, or a related field.

Learn about software engineers, what they do, what they need to know, and how you can begin your career exploration with the software engineer job description below. Afterward, consider trying the IBM DevOps and Software Engineering Professional Certificate to start building skills in microservices, REST APIs, and CI/CD.



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Tools, YAML, Cloud-Native Computing, Continuous Delivery, Jenkins, Infrastructure as Code (IaC), GitHub, Automation, Cloud Security, Cloud Computing, Infrastructure As A Service (IaaS), Cloud Storage, Data Storage Technologies, IBM Cloud, Cloud Computing Architecture, Data Storage, Cloud Services, Cloud Development, Public Cloud, Cloud Engineering, Cloud Platforms, Cross-Functional Collaboration, Performance Measurement, Culture Transformation, Organizational Structure, Performance Metric, Scrum (Software Development), Agile Methodology, Serverless Computing, API Gateway, Cloud Applications, Open Source Technology, Version Control, Software Architecture, Software Development Tools, Front-End Web Development, Unified Modeling Language, Software Development Methodologies, Back-End Web Development, Computer Programming Tools, Full-Stack Web Development, Software Development, Software Design, Development Environment, Software Design Patterns, NumPy, Data Analysis, Secure Coding, Application Security, Vulnerability Scanning, DevSecOps, Security Testing, Threat Modeling, Application Performance Management, System Monitoring, Continuous Monitoring, Prometheus (Software), Dashboard Creation, Interactive Data Visualization, Backlogs, Test Script Development, Kanban Principles, Sprint Retrospectives, Agile Project Management, Agile Software Development, Sprint Planning, Team Oriented, Workflow Management, Lean Methodologies

Software engineer job description: Core responsibilities

Software engineers design, develop, test, and maintain software for end users. Software engineers do more than write code. They ensure that software meets industry needs and is user-friendly.

Software engineers design, develop, test, and maintain software for end users. They do more than write code; they ensure that software meets industry needs and is user-friendly. To achieve this, they follow the software development life cycle (SDLC), a structured process that guides the creation and maintenance of software. This helps ensure the software is developed efficiently, meets quality standards, and remains adaptable to future needs.

The software development life cycle looks like this:

- **Plan:** Engineers define a project's scope, requirements, and timeline.
- **Design:** They propose software architecture, data models, and interfaces.
- **Build:** They write, update, optimize, and integrate code into software applications.
- **Test:** They put software through manual tests, test cases, and QA automation.
- **Deploy:** They release software and training materials to end users.
- **Maintain:** They debug the software, improve UX design, and make updates.

Software engineers often use this process to design a wide variety of software for:

- Computer games
- Operating systems

- Business applications
- Middleware
- Network control systems

Essential skills for software engineers

Software engineers are in high demand in fields such as:

- Aerospace
- Aviation
- Finance
- Cybersecurity
- Data analysis
- Manufacturing
- Health care
- Hospitality
- Insurance
- Marketing

Some software engineers have a university degree. Others attain their skills via coding boot camps or online courses. In any case, key skills for software engineers include:

Coding

While software engineers do more than write code, they need to know how to code to troubleshoot issues with the software they develop. Common coding and computer languages include:

- Python
- C
- C++
- C#
- Java
- JavaScript

- Ruby
- HTML
- CSS
- R
- SQL
- Go
- Rust

The programming languages you learn will depend on your particular software engineering focus.

Read more: Learning How to Code: a Beginner's Guide

[Highlight] You'll also want to be familiar with Agile methodologies. In contrast to the planning and completing of large-scale projects, Agile frameworks have you work in smaller increments, known as sprints, which last from one to four weeks. This shortens the DevOps life cycle. DevOps (comprised of the words development and operations) refers to the coordination of specialties, technologies, people, and previously siloed goals.

This type of corporate unification can result in an improved customer experience, more reliable product development, and the quicker attainment of business goals.

Debugging

Debugging is the process of locating and correcting errors, also known as bugs, in a software's source code. Bugs cause software to malfunction, so software engineers must know how to examine code painstakingly and rectify errors in it carefully.

Common types of software errors include:

- Syntax
- Semantics
- Logic
- Runtime

Software engineers need to have knowledge of tools and testing methods used to test problematic software, such as:

- Unit testing
- Integration testing
- Manual testing

- Automation testing



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Object-oriented programming

Before pursuing software engineering, you may want to familiarize yourself with object-oriented programming (OOP) and object-oriented design (OOD). Key concepts in this realm include:

- **Classes:** Simple code blueprints you can reuse representing abstract ideas with shared attributes (e.g., dogs)
- **Objects:** Specifics derived from abstract classes (e.g., my dog)
- **Encapsulation:** The fact that only some information is available to a broader viewership, while the most important information is contained within an object

- **Inheritance:** When one class extends its features to another, in a sort of parent-child relationship or hierarchy, supporting reusability
- **Polymorphism:** When objects share behaviors in common via method overriding or overloading

Database management

As a software engineer, you must understand how to access and update data quickly and easily. Structured query language (SQL) is the most common method. Important SQL-based tasks include:

- Creating databases, tables, and other structural schemas
- Adding, deleting, and updating data in databases
- Querying tables to filter and retrieve specific data sets
- Combining data from separate tables into one
- Grouping SQL statements to use again
- Managing user permissions and access

Nontechnical skills

To succeed as a software engineer, you'll need a variety of nontechnical workplace skills in addition to the technical skills above. Such skills include:

- Written and verbal communication
- Presentation and visualization
- Active listening and empathy
- Teamwork and leadership
- Adaptability and flexibility

Qualifications and experience requirements

The basic qualification to be a software engineer is a bachelor's degree in software engineering, computer science, mathematics, or a related field. Other qualifications and experience requirements vary based on the type of role you choose.

Types of software engineering roles

If you're looking to get into software engineering, you can find a variety of software engineering roles.

Front-end developer

Front-end developers are concerned with website design and functionality. Users see and interact with a variety of front-end application elements, such as:

- Graphics
- Buttons
- Checkboxes
- Text messages

Front-end developers create a graphical user interface (GUI), the part of a website that users interact with. The three main computer languages they use to do this are:

- HTML
- CSS
- JavaScript

To become a front-end developer, you'll typically need to have at least a bachelor's degree in information technology, computer science, or software engineering. According to Glassdoor, the estimated median total pay for a front-end developer in the US is $97,000 per year. This figure includes base salary and additional pay, which may represent profit-sharing, commissions, bonuses, or other compensation [1].

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Skills you'll build:

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Back-end developer

Back-end developers work on the functionality of a web application. This is also called the “server side” of development. They are less focused on user experience than front-end developers.

Back-end development involves an understanding of various protocols, including:

- Concurrency
- Parallelism
- Caching
- Data duplication

Back-end developers support front-end developers by making code intelligible to front-end applications. You may analyze data, manage application programming interfaces (APIs), and create databases.

Back-end developers usually need a bachelor’s degree in computer science, software engineering, information technology, or a related discipline. According to Glassdoor, the estimated median total pay for a back-end developer in the US is $119,000 per year. This figure includes base salary and additional pay, which may represent profit-sharing, commissions, bonuses, or other compensation [2].

Full-stack developer

Full-stack developers combine the responsibilities of front- and back-end developers. In this role, you use your knowledge to create software from scratch and work to make sure it interfaces well with users.

In addition to typical front-end competency with HTML, CSS, and JavaScript, full-stack developers must also be proficient with:

- NodeJS
- ExpressJS
- Django
- PHP
- MySQL
- PostgreSQL
- MongoDB

Full-stack developers usually need at least a bachelor's degree in a discipline such as computer science, software engineering, information technology, system administration, computer engineering, or web development.

According to Glassdoor, the estimated median total pay for a full-stack developer in the US is $119,000 per year. This figure includes base salary and additional pay, which may represent profit-sharing, commissions, bonuses, or other compensation [3].

Mobile app developer

Mobile app developers create software that can run on mobile devices. These apps usually run on either the Android or the iOS operating platforms.

Mobile app developers work on apps for a variety of devices, such as:

- Smartphones
- Tablets
- Digital assistants

They mainly use these computer languages for their work:

- Java
- Swift
- C#
- HTML5

In the fast-changing world of mobile devices, mobile app developers need to be agile and adaptable. You'll need to have the kind of analytical mindset that can focus on and address software issues. You also need to be creative and interested in finding inventive ways of solving problems.

Mobile app developers usually have a bachelor's degree in computer science, computer programming, or a related field. According to Glassdoor, the estimated median total pay for a mobile app developer in the US is $126,000 per year. This figure includes base salary and additional pay, which may represent profit-sharing, commissions, bonuses, or other compensation [4].

DevOps engineer

DevOps engineers help connect previously siloed departments and direct them toward a unified goal. In other words, they manage a multi-pronged software development process. They work to optimize and sometimes automate parts of this process.

As a DevOps engineer, you foster communication and collaboration among employees. You'll manage system infrastructure and data, develop automation and integration protocols, perform quality assurance, and initiate security protocols.

DevOps engineers often have at least a bachelor's degree in computer science. You will generally need further experience with scripting, programming, automation, cloud technology, and Linux. According to Glassdoor, the estimated median total pay for a DevOps engineer in the US is $144,000 per year. This figure includes base salary and additional pay, which may represent profit-sharing, commissions, bonuses, or other compensation [5].

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